

Automated Conveyor Belt with Object Detection

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Project Status: Simulation Phase Complete

1. Project Overview

This project presents an automated conveyor belt system designed for high-efficiency industrial manufacturing and packaging environments. The system leverages integrated sensing and control logic to streamline workflow by identifying objects in real-time.

The core architecture utilizes:

- **Arduino Uno** for central processing and logic control.
- **HC-SR04 Ultrasonic Sensor** for precise object detection through sonar.
- **L293D Motor Driver IC** to manage the physical movement of the belt via high-current DC motor control.

The system is programmed to halt the conveyor belt immediately upon detecting an item within a predefined range. This pause allows for necessary industrial processing—such as sorting, scanning, or packaging—before the belt automatically resumes its operation.

 **Project Simulation:** [Automated Conveyor Belt Simulation](#)

2. Bill of Materials (BOM)

Component Name	Quantity	Purpose in Circuit	Technical Specification
Arduino Uno R3	1	System microcontroller	ATmega328P, 5V operating voltage
L293D Motor Driver IC	1	High-current motor drive	Quadruple Half-H Drivers, 600mA/ch
HC-SR04 Sensor	1	Ultrasonic object detection	2cm–400cm range, 40kHz freq
DC Gear Motor	1	Mechanical belt rotation	3V–9V operating range
9V Alkaline Battery	1	External motor power source	Standard 9V PP3 block

3. System Architecture

The following block diagram illustrates the flow of control and feedback within the automated system.

Control Path:

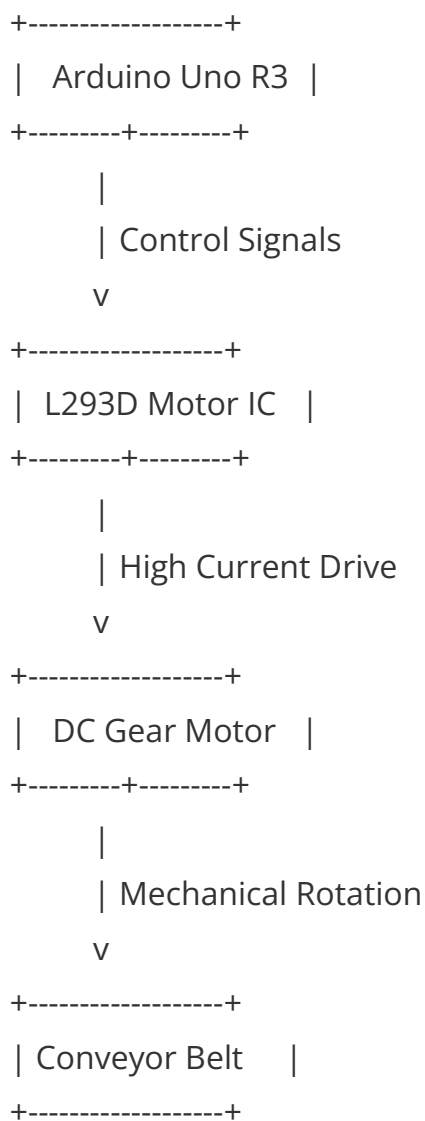
- Arduino Uno R3:** Processes sensor data and sends logic signals.
- L293D Motor Driver:** Receives low-current logic and switches high-current battery power.

3. **DC Gear Motor:** Translates electrical energy into mechanical belt movement.

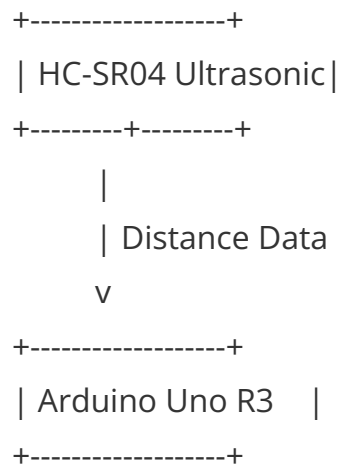
Feedback Loop:

- **HC-SR04 Ultrasonic Sensor:** Continuously monitors for physical obstructions.
- **Distance Data:** Returns pulse-width signals to the Arduino to determine object proximity.

5. BLOCK DIAGRAM



Sensor Feedback:



5. Technical Implementation

The system logic is governed by two primary functions that manage the state of the conveyor belt.

Core Control Functions

```
void runBelt() {  
  
    digitalWrite(motorIn1, HIGH);  
  
    digitalWrite(motorIn2, LOW);  
  
    analogWrite(motorEnable, 200); // Approximately 80% duty cycle  
  
}  
  
void stopBelt() {  
  
    digitalWrite(motorIn1, LOW);
```

```
digitalWrite(motorIn2, LOW);  
  
analogWrite(motorEnable, 0); // Motor off  
  
}
```

6. Functional Explanation

Object Distance Measurement

The HC-SR04 sensor operates on the Sonar principle. It emits an ultrasonic pulse and measures the time taken for the echo to return using the `pulseIn()` function. The distance is then calculated based on the speed of sound, allowing the Arduino to trigger the `stopBelt()` function when an object is detected within range.

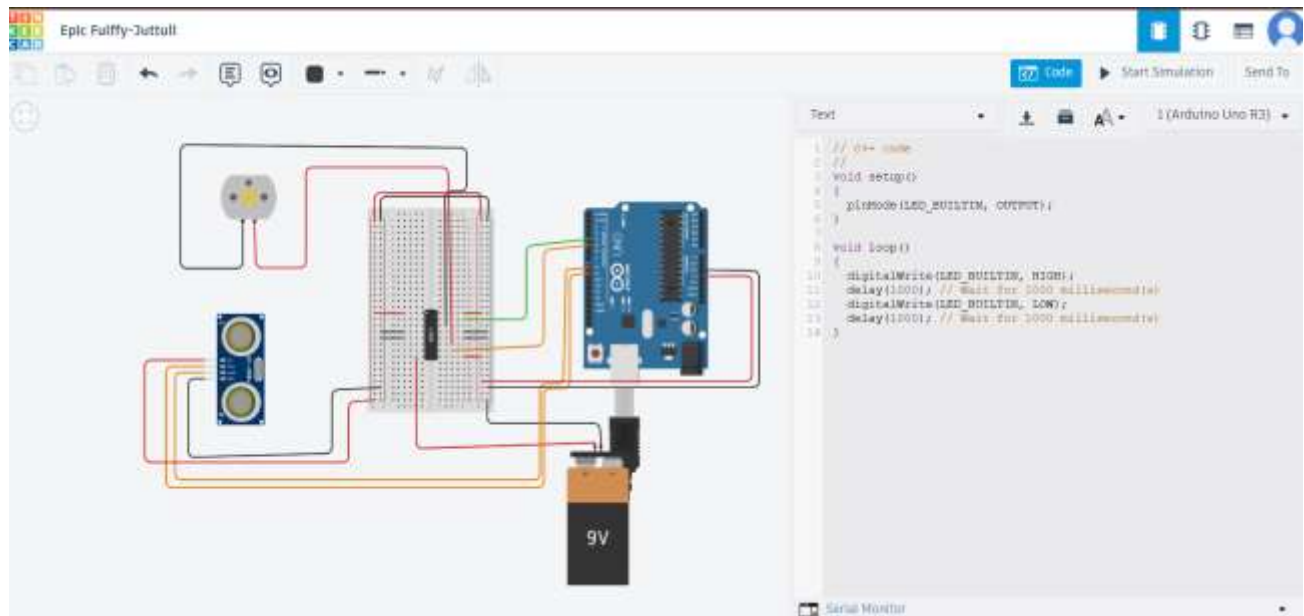
Motor Driver Power Conversion

Because an Arduino cannot provide sufficient current to power a DC gear motor directly, the L293D Motor Driver acts as an H-Bridge interface. It allows the Arduino's 5V logic signals to control the 9V power path from the external battery. Pulse Width Modulation (PWM) is utilized on the Enable pin to maintain consistent belt speed.

7. Industrial Applications

- **Manufacturing:** Precise positioning for automated assembly components.
 - **Logistics:** Smart sorting and routing in warehouse fulfillment centers.
 - **Quality Control:** Automated halting for visual inspection or scanning.
 - **Education:** A foundational model for teaching robotics and mechatronics.
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8. ScreenShort of Thinkercad Simulation



9. Circuit Diagram

